

SAMBIT MISHRA

Ph.D. Student, Electrical & Computer Engineering | University of Southern California

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Research Interests

Causal inference and probabilistic graphical models, with emphasis on identifiability theory and scalable continuous-optimization methods for causal discovery. Interested in efficient discovery algorithms for identifiable DAGs, variance bounds for non-/partially-identifiable cases, and high-dimensional, mixed-distribution data.

Education

University of Southern California (USC)

Los Angeles, CA, USA

Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Urbashi Mitra

Aug 2025 – Present

GPA: 4.00/4.00

Key Coursework: Probability Theory, Linear Algebra, Supervised Machine Learning, Inference & Estimation Theory.

Indian Institute of Technology (IIT) Bhubaneswar

Bhubaneswar, Odisha, India

B.Tech. (Hons.) in Electronics and Communication Engineering; Advisor: Dr. Soumya P. Dash

2021 – 2025

GPA: 9.14/10.00

Publications

- [1] **S. Mishra** and U. Mitra, “Causal Discovery in Equal Variance Linear Gaussian DAGs via SURE-Tuned Ridge Regression,” *arXiv preprint arXiv:2608.17132*, 2026
- [2] C. Peng, **S. Mishra**, U. Mitra, “Learning to Intervene: Optimized Soft Intervention Selection for Causal Discovery,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Barcelona, Spain, 2026, pp. 6196–6200.
- [3] **S. Mishra**, S. P. Dash and G. C. Alexandropoulos, “SER-Optimized Multi-Level ASK Modulations for RIS-Assisted Communications With Energy- and Sign-Based Noncoherent Reception,” in *IEEE Transactions on Green Communications and Networking*, vol. 10, pp. 1433–1445, 2026.
- [4] **S. Mishra** and S. P. Dash, “Error Analysis With Optimal Receiver and Multi-Level ASK for RIS-Assisted Noncoherent Wireless System,” in *IEEE Wireless Communications Letters*, vol. 15, pp. 300–304, 2026.

Research Experience

Graduate Research Assistant

Los Angeles, CA, USA

Ming Hsieh Department of Electrical and Computer Engineering, USC

Aug 2025 – Present

- Developed continuous optimization methods for scalable DAG estimation using differentiable constraints to reduce iterations and enable large-scale batching.
- Designed a soft-intervention selection framework to improve causal discovery efficiency and lower experimental costs.
- Studied identifiability of causal graphs under finite samples and latent variable settings.
- Built GPU-accelerated structure learning pipelines in PyTorch using Adam and L-BFGS for non-convex optimization.

Undergraduate Researcher

Bhubaneswar, Odisha, India

School of Electrical and Computer Sciences, IIT Bhubaneswar

Aug 2024 – May 2025

- Developed SER-optimized modulation schemes for RIS-assisted noncoherent communication systems.
- Conducted error analysis for multi-level ASK modulations, optimizing energy efficiency.

Technical Skills

- **Causal Inference:** Causal Discovery, Bayesian Networks, Structural Equation Models.
- **Optimization:** Gradient-based solvers (Adam, RMSProp, L-BFGS), Convex Optimization, Bilevel Optimization.
- **Programming & Tools:** Python (PyTorch, NumPy, SciPy), C++, CUDA, MATLAB, Git, LaTeX.

Industry Experience

NVIDIA Corporation

Bengaluru, Karnataka, India

ASIC Engineering Intern, CHI VIP Team

May 2024 – July 2024

Verified CHI protocol compliance for CPU/GPU IPs using SystemVerilog and Perl automation; ensured interoperability between internal and external verification IPs.

Awards & Academic Distinctions

- B.Tech. Project Proficiency Award — Best B.Tech Thesis, ECE Department, IIT Bhubaneswar (2024–25)
- All India Rank 4000 of ~1,000,000 candidates, JEE Advanced 2021
- National Talent Search Scholarship — top 2000 of ~1,200,000 candidates, NTSE 2019, Government of India